# Report on CNN Model

**Project:** *Gender and Age Prediction*
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**Dataset Preparation**

The dataset consists of face images with given gender and age labels.

- A custom **PyTorch** Dataset class was developed for loading images and assigning labels.
- The age labels were normalized for better learning using both min–max scaling.

**Preprocessing**

To ensure consistency and robust training:

- Resized to 224×224.
- Applied transforms such as:
  - ToTensor() (to convert images to PyTorch tensors)
  - Normalization (for stable gradient updates)

**Data Splitting**

- The data was divided into 80% - 20% validation split.
- This allowed proper monitoring of learning progress and validation loss during training.

**Initial Architecture**

- Single CNN model was used to predict both gender and age together.
- 3 Convolutional layers followed by ReLU activation and MaxPooling.
- FC (Dense) layers for the joint estimation of gender and age (classification-regression, respectively).

**Training Setup**

- Gender loss: CrossEntropyLoss() (classification)
- Age loss: MSELoss() (regression)
- Both losses were combined with equal weight (≈50–50).
- Initial training was done for 2 epochs for testing purposes.

However, the model learned gender better than age:

- Gender accuracy: nearly 75%
- Age $R^2$ score: ≈ –100 (poor performance)

**Model Improvements**

Seeded the models

- Two separate CNNs were created: one for gender and another for age prediction. This improved learning specialization.
- First normalised age by dividing by 100. Later switched to z-score normalization, using mean and standard deviation.

  This significantly stabilized the regression learning.

Added extra convolutional and dense layers

- Increased model depth and capacity.
- Made Fully connected layer dense for age
- Added dropout layers to prevent overfitting.

**Improved preprocessing**

Rotation, flipping, and color augmentation were included to diversify the training data.

Results

| Model | Accuracy (Gender) | $R^2$ Score (Age) |
|---|---|---|
| Initial Combined Model | 75% | –100 |
| Separated Models | 76% | 0.10 |
| Final Improved Models | 78.6% | 0.30 |